Dynamic Programming

Part 2: Exercise 1 & 2 (week 11 & 12)

Mikkel Reich

University of Copenhagen, Department of Economics

Plan for today

- Structural estimation of Harold Zurcher
- Nested Fixed Point Algorithm (NFXP)

Part 2: Guide

- Last week, we did `00_intro`, solving Harold Zurcher
- This and next week (week 11 + 12), we work in `01_02_NFXP`
  - Week 11: Get to know NFXP and do `01_NFXP.ipynb` (more look through code than code up')
  - Week 12: Simulate data (+ estimate) in `02_NFXP_simulate.ipynb`
- Last week of part 2, we work in `03_NPL` and do `03_NPL.ipynb`
- MANY extra files / exercises that I don't expect you to do, e.g:
  - `03_NFXP_equilibrium.ipynb` in `01_02_NFXP`
  - Folder `04_MPEC`
- Can definitely be relevant for project → Resources for you
- Recommendation: Bertels youtube + DSE youtube

Last week: VFI algorithm with taste-shocks

1. Make arbitrary guess on $\bar{V}^{guess}(x)$
2. Compute next-period expected integrated value function (expectation is over future states x')

$$EV(x,d) = \int_X \bar{V}^{guess}(x')p(x'|x,d)dx'$$

3. Given $EV(x,d)$, compute integrated value function (expectation is over taste-shocks ϵ)

$$\bar{V}(x) = \int_\Omega \left[\max_{d\in\{0,1\}} \{u(x,d) + \epsilon(d) + \beta EV(x,d)\}\right] d\epsilon$$

4. Check convergence $\max\{|\bar{V}(x) - \bar{V}^{guess}(x)|\}$
5. Update $\bar{V}^{guess}(x)$ with $\bar{V}(x)$
6. Keep iterating and check convergence below some threshold

Poly-Algorithm: VFI + Newton–Kantorovich (NK)

VFI: Find fixed point in integrated value function operator

$$\bar{V}(x) = \bar{\Gamma}(\bar{V})(x) \Leftrightarrow (I - \bar{\Gamma})(\bar{V})(x) = 0$$

Generally, VFI is

1. fast in the beginning since we quickly get "better guesses"
2. slow later on with small updates (linear convergence)

Idea: When VFI (/SA) is slow $\rightarrow$ Do NK-iterations

$$\bar{V}_{k+1} = \bar{V}_k - (I - \bar{\Gamma}')^{-1}(I - \bar{\Gamma})(\bar{V}_k)$$

NK: (+) quadratic convergence (-) requires good initial guess

For NK, we need Frechet derivative of Bellman operator: $\bar{\Gamma}'$

NK: Frechet Derivative

Bellman operator $\bar{\Gamma}(\bar{V})$ is an operator: Function of function

In code: $\bar{\Gamma} : \mathbb{R}^n \to \mathbb{R}^n$, where n is number of points in grid for x

$\to$ Think of abstract Frechet derivative as Jacobian matrix

$$\bar{\Gamma}(\bar{V}) = \ln\left(\overbrace{\exp(u(x,0) + \beta P_0^M \bar{V}) + \exp(u(x,1) + \beta P_1^M \bar{V}))}^{=S}\right)$$

$$\bar{\Gamma}'(\bar{V}) = \text{diag}\left(\frac{\exp(u(x,0) + \beta P_0^M \bar{V})}{S}\right)\beta P_0^M$$

$$+ \text{diag}\left(\frac{\exp(u(x,0) + \beta P_0^M \bar{V})}{S}\right)\beta P_1^M$$

$$= \beta(\text{diag}(P(d=0|x))P_0^M + \text{diag}(P(d=1|x))P_1^M)$$

, where $P(d=0|x)$ and $P(d=1|x)$ are choice probabilities

and P_0^M and P_1^M are Markov (transition) matrices given choices

What does VFI give us? How can we use it?

We get conditional choice probabilities: $\{P(d = \text{keep} = 0|x, \theta)\}_{x \in X}$, given (structural) parameters of model θ

Think of *not changing engine* as (binary) stochastic variable

Binary $\rightarrow$ Bernoulli distribution $\quad \Rightarrow d = 0|x \sim Ber(P(d = 0|x, \theta))$

Could do MLE and think of all observations $\{d_i, x_i\}_{i=1}^N$ as i.i.d., define $\ell(\theta) = \log\left(L(\theta|\{d_i, x_i\}_{i=1}^N\right)$

$$\ell(\theta) = \sum_{i=1}^{N}(1 - d_i) * \log(P(d_i = 0|x_i, \theta)) + d_i * \log(1 - P(d_i = 0|x_i, \theta))$$

Could add log-likelihood from $\{\Delta x_i\}_{i=1}^N$, which in our "grid-world" is multinominally distributed with $Supp(\Delta x) = \{0, 1, 2, 3, 4\}$

Some none-lethal derivatives 1/2

def' $P_i(\theta) = P(d_i = 0|x_i, \theta)$ and compute score

$$s(\theta) = \sum_{i=1}^{N}(1-d_i)\frac{1}{P_i(\theta)}\frac{\partial P_i}{\partial \theta} - d_i\frac{1}{1-P_i(\theta)}\frac{\partial P_i}{\partial \theta}$$

$$= \sum_{i=1}^{N}\frac{\partial P_i}{\partial \theta}\frac{(1-d_i)(1-P_i(\theta)) - d_i P_i(\theta)}{P_i(\theta)(1-P_i(\theta))}$$

$$= \sum_{i=1}^{N}\frac{\partial P_i}{\partial \theta}\frac{1-P_i(\theta) - d_i + d_i P_i(\theta) - d_i P_i(\theta)}{P_i(\theta)(1-P_i(\theta))}$$

$$= \sum_{i=1}^{N}\frac{\partial P_i}{\partial \theta}\frac{1-P_i(\theta) - d_i}{P_i(\theta)(1-P_i(\theta))}$$

Use property of logit-formula + def' $v_{d,i} = u(x_i, d_i) + \beta P_d \bar{V}$

$$\frac{\partial P_i}{\partial \theta} = P_i(\theta)(1-P_i(\theta))\left(\frac{\partial v_{0,i}}{\partial \theta} - \frac{\partial v_{1,i}}{\partial \theta}\right)$$

$$\Rightarrow s(\theta) = \frac{1}{N}\sum_{i=1}^{N}(1-P_i(\theta) - d_i)\left(\frac{\partial v_{0,i}}{\partial \theta} - \frac{\partial v_{1,i}}{\partial \theta}\right)$$

Some none-lethal derivatives 2/2

To compute score $\rightarrow$ Need to compute $\frac{\partial v_{0,i}}{\partial \theta}$ and $\frac{\partial v_{1,i}}{\partial \theta}$

$$\frac{\partial v_{d,i}}{\partial \theta} = \frac{\partial u_{d,i}}{\partial \theta} + \beta \left(\frac{\partial P_d^M}{\partial \theta} \bar{V} + P_d^M \frac{\partial \bar{V}}{\partial \theta} \right)$$

w/o likelihood from state trans., we have $\frac{\partial P_d^M}{\partial \theta} = 0$

Implicit function theorem gives us $\frac{\partial \bar{V}}{\partial \theta}$:

$$\bar{V}(\theta) - \bar{\Gamma}(\bar{V}(\theta); \theta) = 0$$

$$\frac{\partial \bar{V}}{\partial \theta} - \frac{\partial \bar{\Gamma}}{\partial \bar{V}} \frac{\partial \bar{V}}{\partial \theta} - \frac{\partial \bar{\Gamma}}{\partial \theta} = 0 \Leftrightarrow \frac{\partial \bar{V}}{\partial \theta} = \left(I - \frac{\partial \bar{\Gamma}}{\partial \bar{V}} \right)^{-1} \frac{\partial \bar{\Gamma}}{\partial \theta}$$

In code: $\frac{\partial \bar{\Gamma}}{\partial \bar{V}}$ is (n, n) array and $\frac{\partial \bar{\Gamma}}{\partial \theta}$ is $(n, n_\theta) \rightarrow \frac{\partial \bar{V}}{\partial \theta}$ is (n, n_θ)

Envelope theorem: Only think of direct effect of θ on $\bar{\Gamma}$

$$\Rightarrow \frac{\partial \bar{\Gamma}}{\partial \theta} = P(d = 0|x) \frac{\partial u_0}{\partial \theta} + P(d = 0|x) \frac{\partial u_1}{\partial \theta}$$

Nested Fixed Point Algorithm (NFXP) for Harold Zurcher

3 Steps:

1. Estimate m (or $\{\Delta x_i\}_{i=1}^N$) as frequencies: $\tilde{p}_j = \frac{\#\{\Delta x_i = j\}}{N}$ for $j = 0, 1, 2, 3$ and $\tilde{p}_4 = 1 - \sum_{j=1}^3 p_j$. Think of this as "qualified guess" on $\{\tilde{p}_j\}_{j=0}^3$
2. Do maximum likelihood estimation for (RC, c) using BHHH ("MLE-optimizer" with hessians "for free")
   2.1 For every log-likelihood evaluation: Find fixed point $\bar{V}_\theta = \bar{\Gamma}_\theta(\bar{V}_\theta)$
3. Do MLE for full parameter set $(RC, c, p_0, p_1, p_2, p_3)$ using $(\hat{RC}, \hat{c})$ and $\{\tilde{p}_j\}_{j=0}^3$ as qualified guesses (again find fixed point)

Get variance using MLE-variance formula, following Information Matrix Equality: $Var(\hat{\theta}) = (s(\theta)^T s(\theta))^{-1}$

Your time to shine!